

Vivian Li

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EDUCATION

Brown University | Providence, RI, USA

Sc.M. in Electrical and Computer Engineering (Research Track)

September 2025 – May 2026

- Recipient of School of Engineering Master's tuition award

Sc.B. in Computer Science (Visual Computing + AI + Design)

September 2021 – May 2025

- Cross-registration at **Massachusetts Institute of Technology (MIT)** and **Rhode Island School of Design (RISD)**

EXPERIENCE

Brown University – Research Assistant, Computational Design Group

September 2025 – *Present*

- First-author work on an end-to-end computational design and fabrication system for programmable surface deformation in embroidered textiles (UIST 2026 Short Paper, to appear).
- Co-authored DicePlay (SIGGRAPH 2026), a grammar-based optimization framework for a modular display made of dice.
- Ongoing research on mixed discrete–continuous optimization for multi-objective garment design.
- Advised by **Prof. Adriana Schulz**.

Walt Disney Animation Studios – Production Technology Intern | Burbank, CA, USA

June 2024 – August 2024

- Designed, implemented, and deployed a C++/Qt UI State Debugger in Presto (Pixar's proprietary animation & rigging software) to visualize complex debugging states and support artist and production workflows.
- First intern project adopted into production pipelines at both Pixar and Walt Disney Animation Studios.

MIT – Research Assistant, Fibers@MIT | Cambridge, MA, USA

January 2024 – June 2024

- Developed a knit-integration technique for embedding fiber-based computation in textiles, enabling fabric-scale sensing and distributed inference; co-authored publication in *Nature* (2025).
- Advised by **Prof. Yoel Fink**.

Tsinghua University – Research Intern, Future Lab | Beijing, China

June 2023 – August 2023

- Led research on a generative pipeline for creating buildable LEGO models and assembly instructions from text and images, investigating 3D generation and LEGO representations.
- Supervised by **Prof. Yingqing Xu**.

Microsoft Research Asia – Internet Graphics Intern | Beijing, China

July 2022 – August 2022

- Researched lighting and virtual environment design for *Project VirtualCube*, an immersive telepresence system that renders remote participants in real time while preserving 3D geometry, texture, and mutual eye gaze.
- Supervised by **Dr. Xin Tong**.

Brown University – Undergraduate Research Assistant, Dash | Providence, RI, USA

Jan. 2022 – Jan. 2023

- Developed a full-stack multicolumn notetaking feature for Dash, a collaborative hypermedia web application supporting scalable document-based knowledge workflows, with document linking and markup.
- Advised by **Prof. Andries van Dam**.

PUBLICATIONS

StitchSwitch: Programmable Surface Deformation and Bistability in Embroidered Textiles

ACM UIST, 2026 (To Appear)

Vivian Li, Milin Kodnongbua, Heather Robertson, Yiyue Luo, Adriana Schulz

DicePlay: A Modular Canvas for Physical Image Composition

ACM SIGGRAPH, 2026

Milin Kodnongbua*, Zihan Jack Zhang*, Shishi Xiao*, **Vivian Li**, Heather Robertson, Rulin Chen, David Laidlaw, Adriana Schulz

Perceptual Sculptures: Computational Design of View-Dependent Hanging Art

ACM Symposium of Computational Fabrication (SCF), 2026 (Accepted)

Christopher Mok, **Vivian Li**, Ashton Glover*, Chaewon Bae*, Adriana Schulz, Milin Kodnongbua

Computational Design of Tunnel Books

ACM Symposium of Computational Fabrication (SCF), 2026 (Accepted)

Zihan Jack Zhang*, Apoorva Talwalkar*, **Vivian Li**, Jaclyn Cohen, Nicole Sanchez-Soto, James Tompkin, Adriana Schulz

A Single-fibre Computer Enables Textile Networks and Distributed Inference

Nature, 2025

Nikhil Gupta, Henry Cheung, Syamantak Payra, Gabriel Loke, Jenny Li, Yongyi Zhao, Latika Balachander, Ella Son, **Vivian Li**, Samuel Kravitz, Sehar Lohawala, John Joannopoulos and Yoel Fink

TEACHING EXPERIENCE

Head Teaching Assistant, CSCI1953B Computational Design and Fabrication November 2025 – May 2026

- Mentored student final projects, contributing fabrication work that helped advance two projects toward acceptance for publication at ACM SCF 2026 (Perceptual Sculptures, Computational Design of Tunnel Books).
- Led course development and operations for its inaugural undergraduate offering at Brown, coordinating TA activities, creating assignments and rubrics, overseeing grading, and leading a laser-cutting workshop.

TALKS

Invited Speaker at Boston University Graphics Group Meeting

August 2026

SELECTED PROJECTS

Kinetic Pixels | Robotic Swarm Simulation October – December 2025

- Designed a robotic swarm simulation for rendering dynamic physical mosaics using cellular automata and collision-aware path planning.

Aligned | Embedded Electronics Design October 2024 – April 2025

- Built a posture-sensing system for textile integration with a custom PCB, I²C accelerometers, haptic feedback, and real-time machine learning for posture classification.

Color Me Noisy | Advanced Computer Graphics Project March – May 2025

- Reimplemented an example-based rendering pipeline that stylizes 3D animations as temporally coherent hand-colored artwork using multi-scale texture synthesis.

Spirit Oasis | Computer Graphics Project November – December 2024

- Recreated the Spirit Oasis scene from *Avatar: The Last Airbender*, implementing procedural animation, custom mesh deformers, and real-time rendering in Three.js with assets modeled in Blender.

Aerial Reverie | Coordinated Robotics VR Short Film March – May 2024

- Programmed and directed an immersive VR storyworld featuring a dynamic cityscape built by coordinated Crazyflie drone flight paths using Python and ROS.

PawLink | Computational Textiles November – December 2023

- Designed and developed a knitted smart collar integrating fiber-based sensing and on-device machine learning for real-time animal health monitoring.

SKILLS

Languages: C++, Python, JavaScript/TypeScript, Java, HTML/CSS, MATLAB, LaTeX

Software & Frameworks: Qt, React, OpenGL, Arduino, Linux

3D/2D Design: Blender, Maya, Unity, Fusion 360, Marvelous Designer, Adobe Creative Suite, Figma

Fabrication: PCB & Schematic Design, 3D Printing, Laser Cutting, Knitting, Sewing, Embroidery

LEADERSHIP & SERVICE

Co-Founder & Web Lead, SenseScape: *Beyond Boundaries* | NYC, NY, USA October 2023 – May 2025

- Co-organized an innovation conference and exhibition at Microsoft Garage NYC featuring XR installations, live demos, and panel discussions for 200+ attendees. Led development of an immersive 3D exhibition website archiving conference works with future VR accessibility in mind.

Design Lead, Hack@Brown | Providence, RI, USA

September 2023 – March 2025

- Led the design team in developing the website and 3D visual identity for [Hack@Brown 2025](#) and 2024, attracting 500+ participants from across the country each year.

Outreach Lead, Founding Member, Brown/RISD XR Hub | Providence, RI, USA

Sept. 2023 – Dec. 2024

- Founding executive board member who led outreach to industry speakers and partners while organizing interdisciplinary XR events connecting students across Brown and RISD.

MEDIA

RISD News — "[RISD and Brown Students Present Innovation Conference at Microsoft Garage NYC](#)" (2025)

Brown Daily Herald — "[MIT, RISD and Brown researchers develop computers you can wear](#)" (2025)